

## Response Letter with Annotations

### *Citation (APA format, using Zotero or other bibliography software)*

Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science*, 359(6380), 1146–1151. <https://doi.org/10.1126/science.aap9559>

### Annotations

#### 1. Purposes

- a. Motivation: There is a need to understand how falsity spreads online, especially with the advent of new social technologies that facilitate rapid information sharing and large-scale information cascades, which can enable the spread of misinformation
- b. Specific Rationale (several bullet points)
  - Prior studies focused on:
    1. Anecdotal analysis of the spread of false news by the media
    2. Studies are limited to analyses of small, ad hoc samples
    3. Single rumors or isolated events (e.g., disasters)
  - Lack of:
    1. Comparative analysis of true vs false news
    2. Few large-scale empirical investigations of the diffusion of misinformation or its social origins.
  - Need to:
    1. Evaluate differences in the spread of truth and falsity across topics
    2. Examine why false news may spread differently than truth
- c. Research Questions and/or Hypotheses
  - How do truth and falsity diffuse differently?
  - What factors of human judgment explain these differences?
  - Identify IV, DV, mediator, moderator
    1. IV: Veracity of news (true vs false)
    2. DV: Diffusion depth, speed, breadth, and structural virality
    3. Mediators: Novelty and emotions?
    4. Moderators: Topic (e.g., politics, urban legends, science)
  - Identify specific relationships among variables
    1. False news → higher novelty → increased retweeting → greater diffusion
    2. False news → stronger emotional reactions → increased sharing

#### 2. Method

- a. Research Design
  - Computational study
  - Observational big data design

**b. Sample**

- Twitter data from 2006 to 2017
- 126,000 rumor cascades (A rumor cascade begins on Twitter when a user asserts a topic in a tweet, which could include written text, photos, or links to articles online).
- 3 million users spread information more than 4.5 million times
- Verified by 6 fact-checking organizations (snopes.com, politifact.com, factcheck.org, truthorfiction.com, hoax-slayer.com, and urbanlegends.about.com)
- Veracity had been agreed on by these organizations (95–98% agreement)

**c. Data collection**

- All English-language replies to tweets that contained a link to any of the aforementioned websites from 2006 to 2017 and used optical character recognition to extract text from images where needed.
- All original tweets being replied to and the retweets of the original tweet were extracted

**d. Measures**

- **Diffusion metrics:**
  1. Depth: the number of retweet hops from the origin tweet over time, where a hop is a retweet by a new, unique user.
  2. Size: the number of users involved in the cascade over time.
  3. Maximum breadth: the maximum number of users involved in the cascade at any depth.
  4. Structural virality: a measure that interpolates between content spread through a single, large broadcast and that which spreads through multiple generations, with any one individual directly responsible for only a fraction of the total spread.
- **Novelty:**
  1. Randomly selected 5000 users and 25000 tweets
  2. Topic modeling (LDA) (200 topics, trained on 10 million English language tweets)
  3. Information distance metrics
- **Emotion**
  1. National Research Council Canada (140,000 English words) emotion lexicon (e.g., fear, surprise, trust)
  2. 32,000 Twitter hashtags and weighted associations with the same emotions
  3. Removal of stop words and URLs from the reply tweets
  4. Calculated the fraction of words in the tweets that related to each of the eight emotions, creating a vector of emotion weights for each reply that summed to one across the emotions.
- **Control variables:**
  1. Account age
  2. Activity level
  3. Number of followers and followees of the original tweeter,

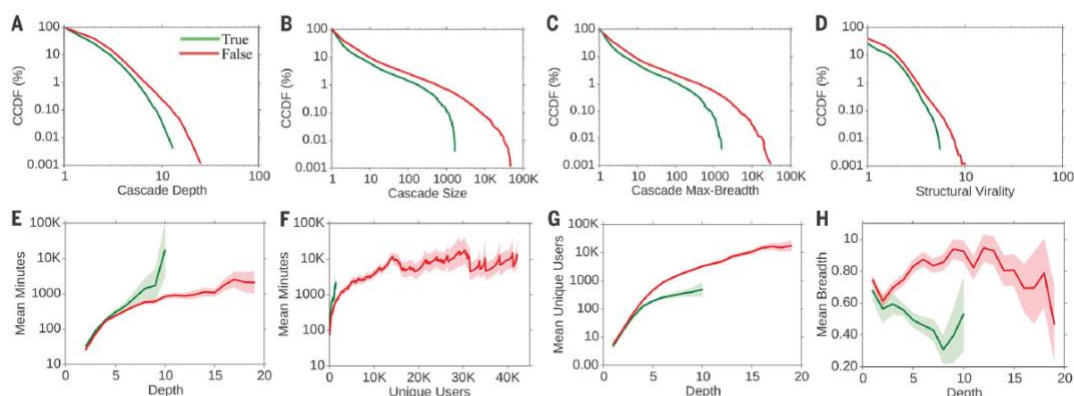
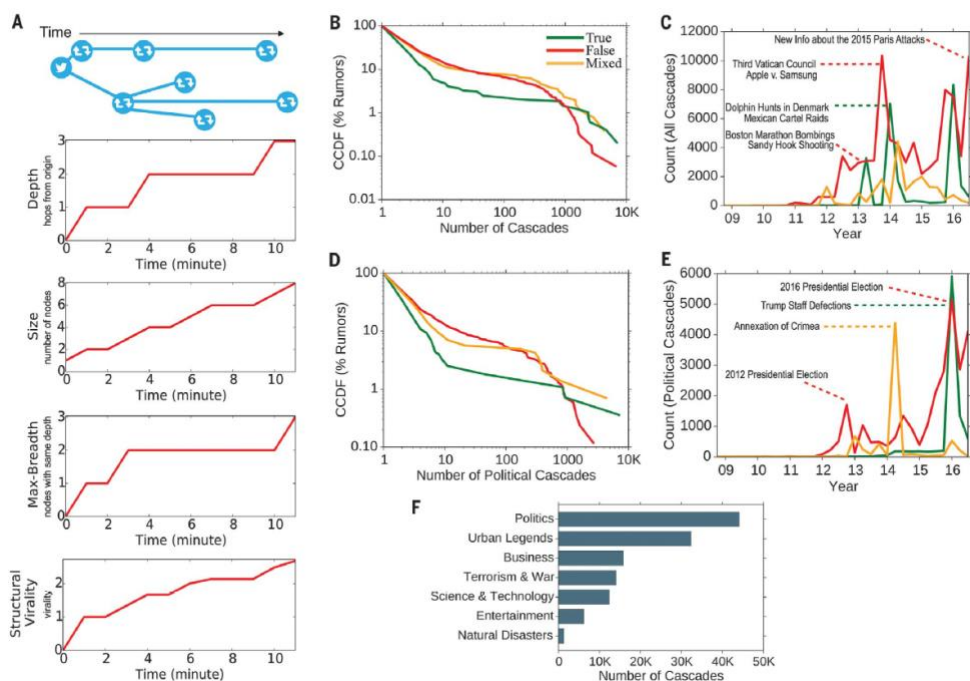
4. Whether the original tweeter was a verified user
- e. Analysis
- Kolmogorov-Smirnov (K-S) tests: used to analyze and compare the diffusion dynamics (speed, depth, and reach) of true versus false rumors
  - Topic modeling (LDA)
  - Logistic Regression: estimated a logistic regression model to determine the likelihood of a rumor being retweeted
  - Robustness checks (bot removal, alternative samples)

### 3. Results and Conclusions

a. Answers to RQs and Hypotheses

- False news diffuses farther, faster, deeper, and more broadly than true news
- False news is 70% more likely to be retweeted
- False news reaches more people (up to 100,000 vs 1,000 for true news)
- The spread of falsehood was aided by its virality, meaning that falsehood did not simply spread through broadcast dynamics but rather through peer-to-peer diffusion characterized by a viral branching process (Fig. 2D).
- False political news traveled deeper (Fig. 3A) and more broadly (Fig. 3C), reached more people (Fig. 3B), and was more viral than any other category of false information.
- Users who spread false news had significantly fewer followers (K-S test = 0.104,  $P \sim 0.0$ ), followed significantly fewer people (K-S test = 0.136,  $P \sim 0.0$ ), were significantly less active on Twitter (K-S test = 0.054,  $P \sim 0.0$ ), were verified significantly less often (K-S test = 0.004,  $P < 0.001$ ), and had been on Twitter for significantly less time.
- False information was more novel than true information
- False rumors inspired replies expressing greater surprise (K-S test = 0.205,  $P \sim 0.0$ ), corroborating the novelty hypothesis, and greater disgust.
- The truth inspired replies that expressed greater sadness (K-S test = 0.037,  $P \sim 0.0$ ), anticipation (K-S test = 0.038,  $P \sim 0.0$ ), joy (K-S test = 0.061,  $P \sim 0.0$ ), and trust (K-S test = 0.060,  $P \sim 0.0$ ) (Fig. 4, D and F).

b. Include Key Figures/Tables

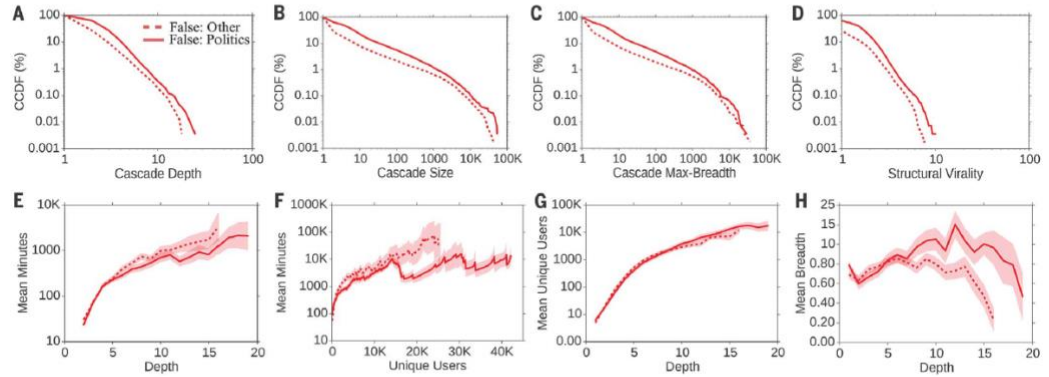


**Fig. 2. Complementary cumulative distribution functions (CCDFs) of true and false rumor cascades.** (A) Depth. (B) Size. (C) Maximum breadth. (D) Structural virality. (E and F) The number of minutes it takes for true and false rumor cascades to reach any (E) depth and (F) number of unique Twitter users. (G) The number of unique Twitter

users reached at every depth and (H) the mean breadth of true and false rumor cascades at every depth. In (H), plot is lognormal. Standard errors were clustered at the rumor level (i.e., cascades belonging to the same rumor were clustered together; see supplementary materials for additional details).

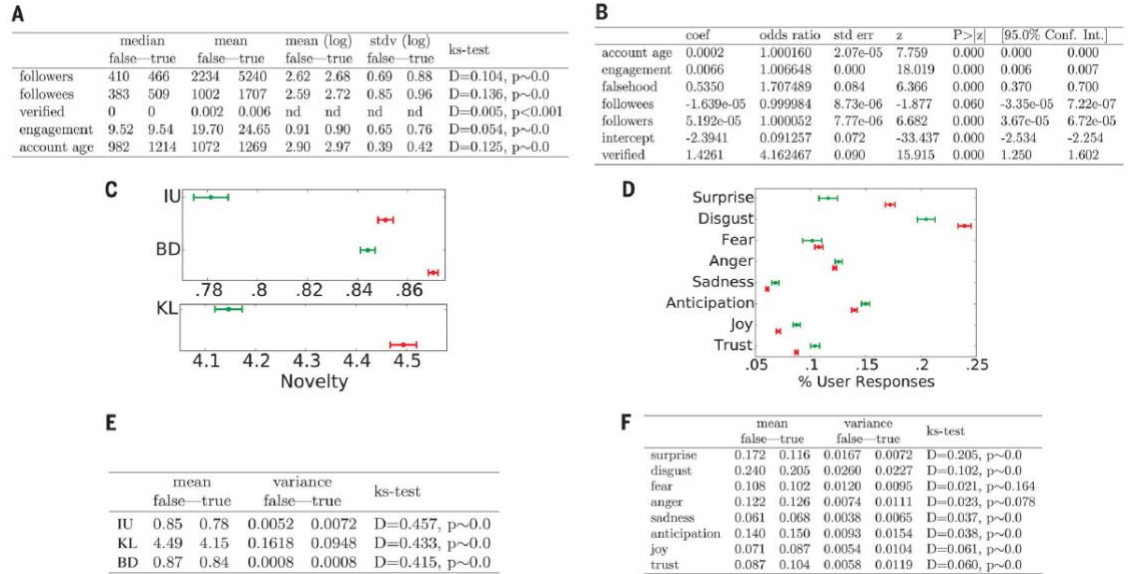
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**Fig. 3. Complementary cumulative distribution functions (CCDFs) of false political and other types of rumor cascades.** (A) Depth. (B) Size. (C) Maximum breadth. (D) Structural virality. (E and F) The number of minutes it takes for false political and other false news cascades to reach any (E) depth and (F) number of unique Twitter users. (G) The number of unique Twitter users reached at every depth and (H) the mean breadth of these false rumor cascades at every depth. In (H), plot is lognormal. Standard errors were clustered at the rumor level.

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### c. Authors' Conclusions

- False rumors spread more persuasively than true rumors
- The greater likelihood of people retweeting falsity more than the truth is what drives the spread of false news, despite network and individual factors that favor the truth.
- Human behavior contributes more to the differential spread of falsity and truth than automated robots do.

## 4. Discussion

### a. Implications

- False news can drive the misallocation of resources during terror attacks and natural disasters, the misalignment of business investments, and misinformed elections.
- Misinformation containment policies should also emphasize behavioral interventions, like labeling and incentives to dissuade the spread of misinformation, rather than focusing exclusively on curtailing bots.

**b. Limitations**

- Observational data that establishes no causal influence. Although we cannot claim that novelty causes retweets or that novelty is the only reason why false news is retweeted more often, we do find that false news is more novel and that novel information is more likely to be retweeted.
- Limited direct measurement of user psychology

**c. Future Directions**

- More research on the behavioral explanations of differences in the diffusion of true and false news is clearly warranted.
- Spread of true and false news online requires more direct interaction with users through interviews, surveys, lab experiments, and even neuroimaging

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## **Response Letter (~600 words)**

### **Assessment of the study**

The study by Vasoughi et al. (2018) is highly relevant in our understanding of misinformation in the digital age. By leveraging a massive dataset of Twitter activity spanning over a decade, the authors investigate how true and false rumors spread on Twitter and find that false information diffuses farther, faster, deeper, and more broadly than true information. The findings challenge the assumption that misinformation is widely spread by bots. Instead, the study shows that human behavior plays a dominant role in the spread of rumors.

One strength of the study is its scale and methodological rigor. The dataset includes approximately 126,000 rumor cascades and millions of users (3 million), making it more robust than earlier studies that focused on smaller samples or relied on event-specific samples. The use of multiple independent fact-checking organizations to classify truth versus falsity strengthens the validity of the key independent variable. The further steps taken by the researchers to reduce selection bias and remove bots add another layer of robustness to the study. The study also integrates several computational techniques, such as topic modeling (LDA) and statistical modeling, to examine diffusion dynamics, novelty, and emotions. This method provides a robust and complex computational design.

Even though computational methods cannot be used to establish causality, the authors did not focus only on the descriptive findings. They went further to explain why false information spreads faster (by measuring novelty and emotions). The findings showed that novelty and emotions cause false rumors to spread farther, faster, deeper, and broader.

### **Weaknesses**

While the authors take several steps to explain why false information spreads faster and farther, the study is observational in nature. Measures of novelty, emotions, and diffusion cannot definitively prove the rationale behind people sharing rumors. The dataset is restricted to rumors that were fact-checked by fact-checking organizations, which may introduce selection bias. Fact-checked stories may differ systematically from the broader universe of online content. I consider this less of a weakness because the authors took further steps to mitigate this limitation.

The focus on Twitter only makes the data limited. Although Twitter is an important platform for information dissemination, it is not a representative of all social media environments. Platforms like Facebook, TikTok, or private messaging apps may exhibit different diffusion dynamics. This affects the generalizability of the study, as the sample is restricted to Twitter users and does not represent the broader population. Not everyone uses Twitter, and those who do may differ in, for example, demographics and media consumption habits from people on other platforms or online. The study does not account for how rumors spread through offline channels, such as face-to-face conversations, traditional media, etc. Due to this, the results may not fully capture the overall dynamics of misinformation diffusion in society. Notwithstanding these limitations, I believe that no single study can capture everything about the spread of misinformation.

### **Other notes**

Overall, this study makes a significant contribution to both communication research and the broader understanding of misinformation. The key insight that humans tend to spread false, novel, and emotional information has important implications for policy and platform design. Efforts to combat misinformation should also consider human behavior and their tendency to share false information. The study was initially difficult to read. Unlike other studies that clearly

separate sections for introduction, methods, and findings, this study takes a different approach.

The findings are embedded within the methods. I only realized that the author tested novelty and emotions when I read further. It seemed like the aspects of novelty and emotions were an afterthought rather than initially planned. Additionally, the findings from this study contradict studies that have examined how rumors or misinformation spread online. Research has shown that less than or equal to 1% of the population is exposed to misinformation.